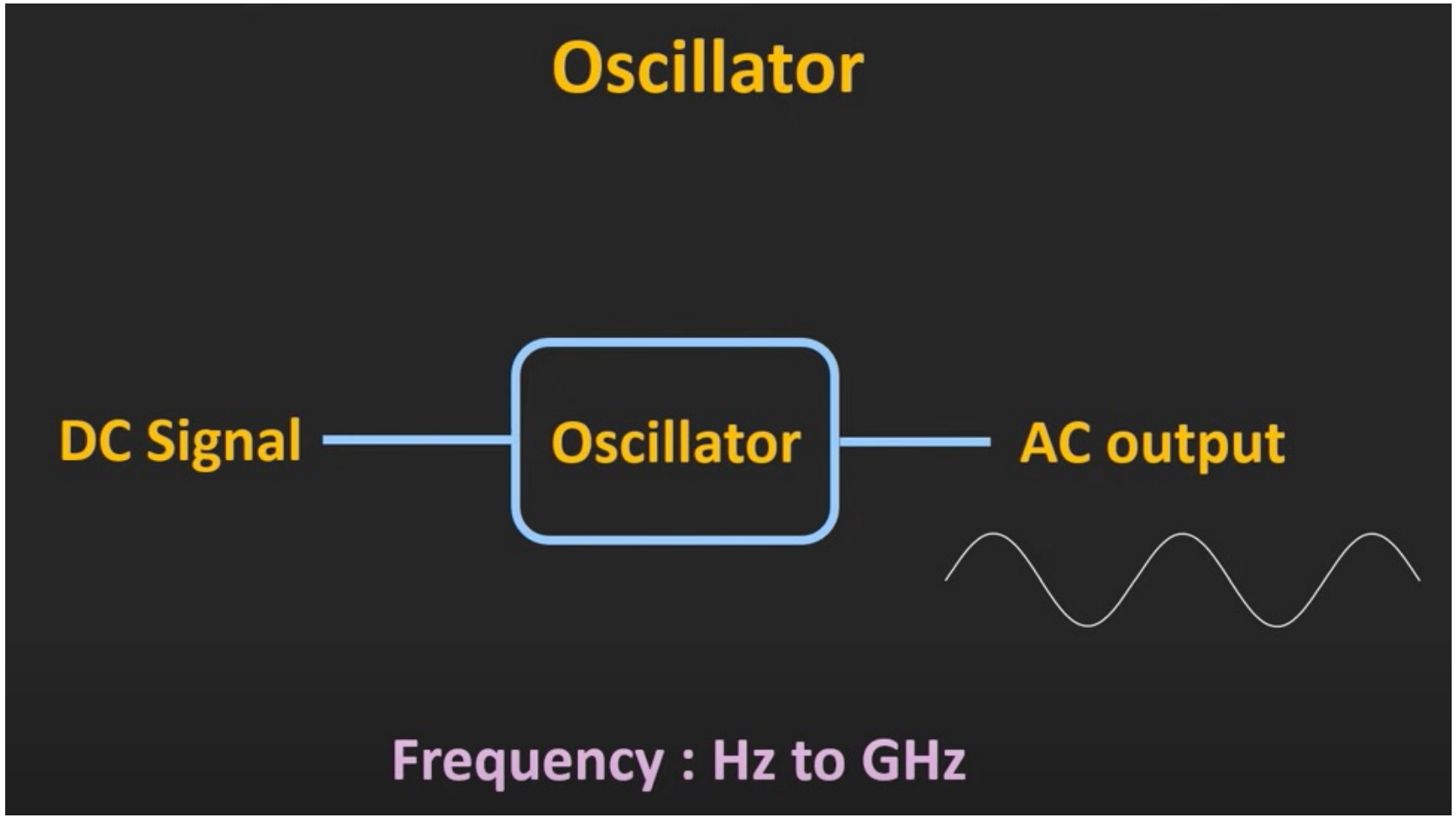
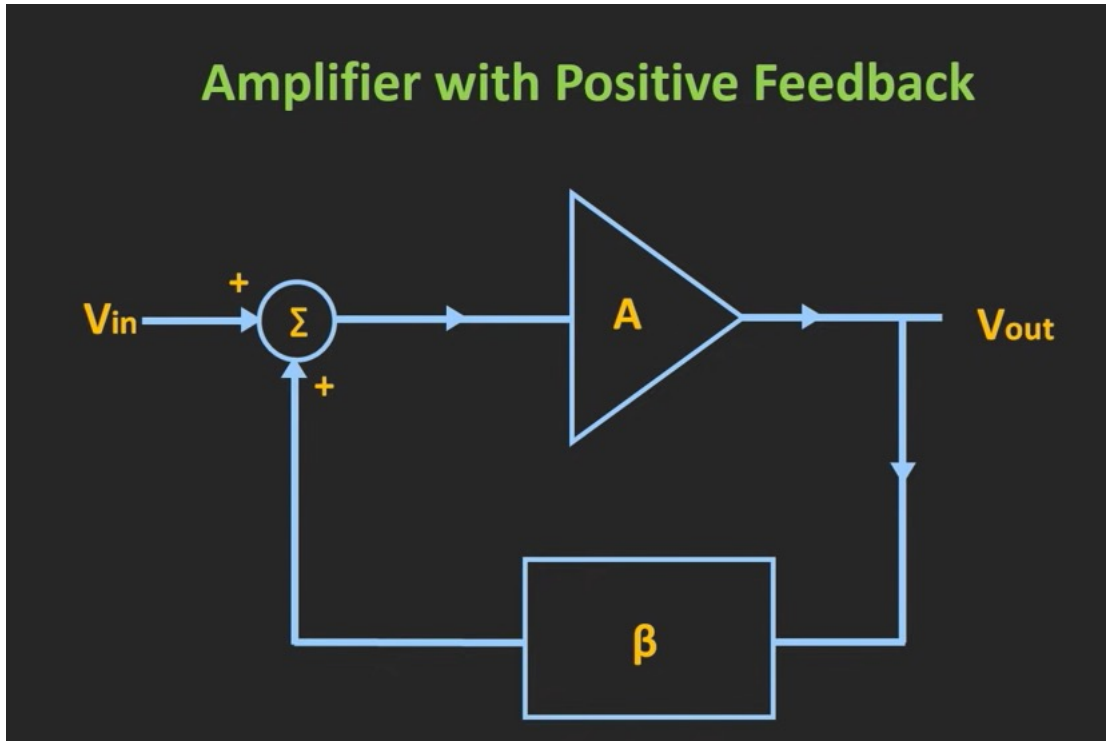
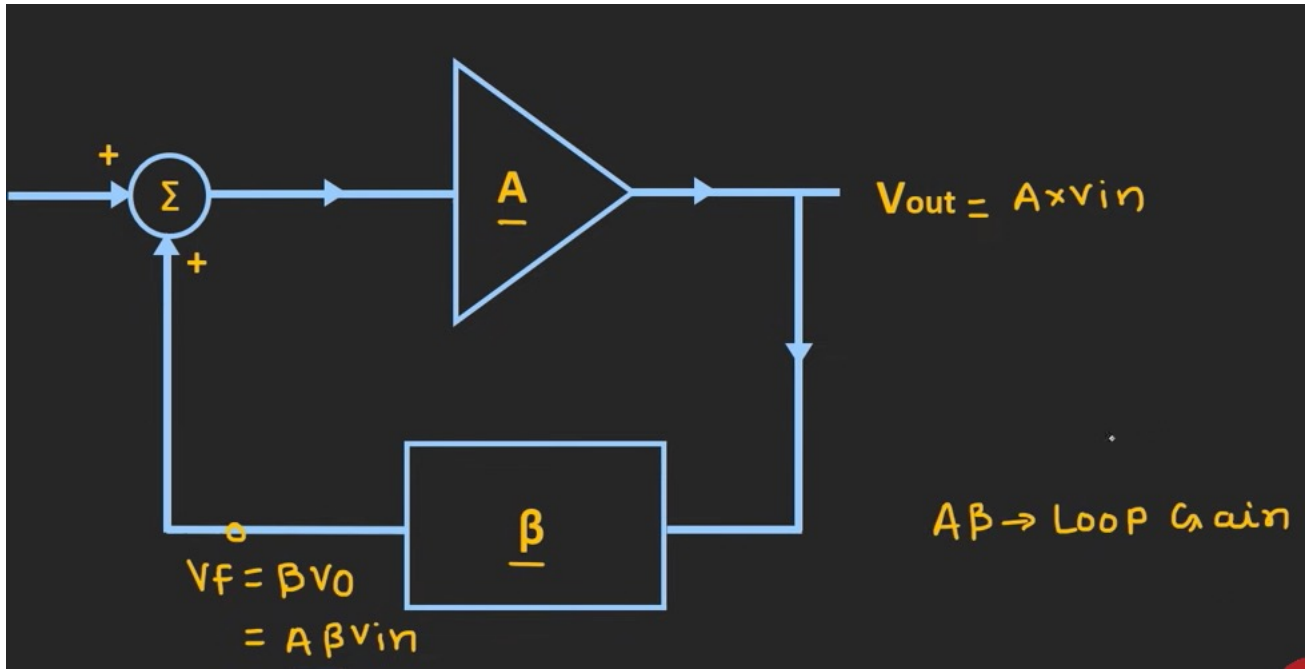
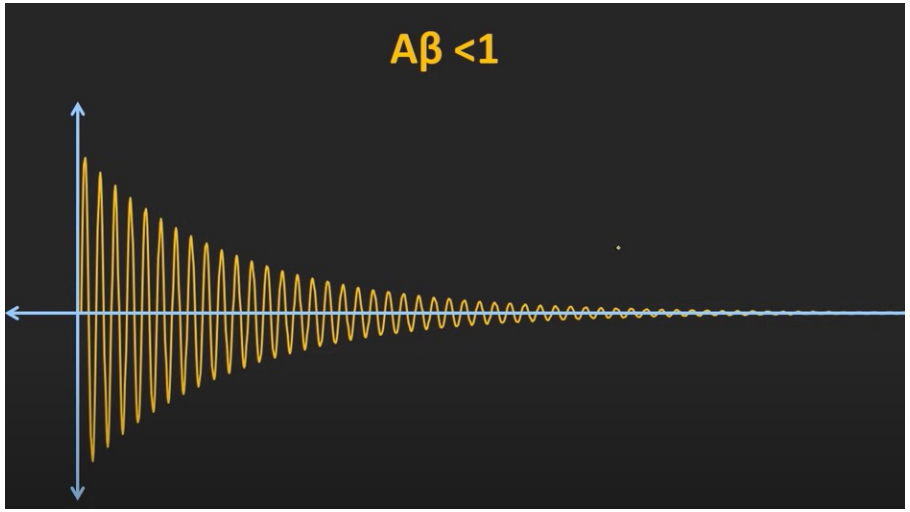
Oscillator

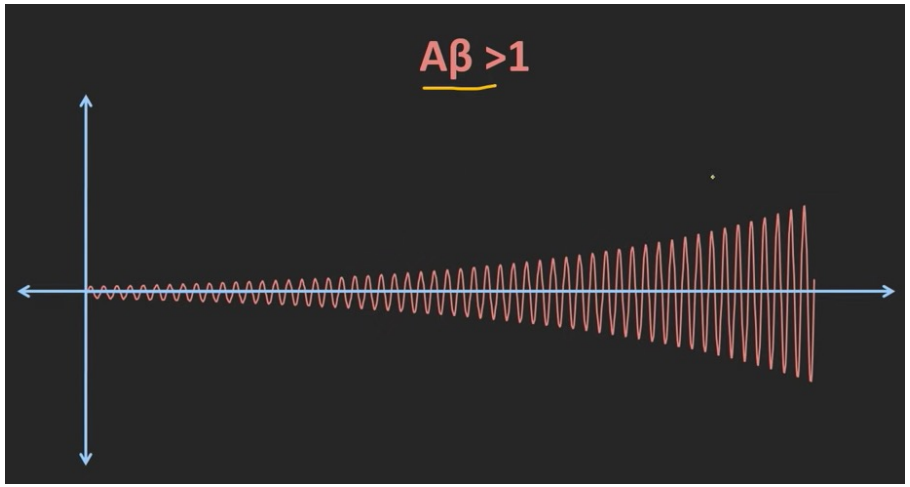
Ginanjar S. Adi


Oscillator
DC Signal
Oscillator
AC output
Frequency : Hz to GHz


Amplifier with Positive Feedback
Vin
+
Σ
+
A
Vout
β


+
Σ
+
A
Vout = A×vin
β
Aβ → Loop Gain
Vf = βVo
= Aβvin

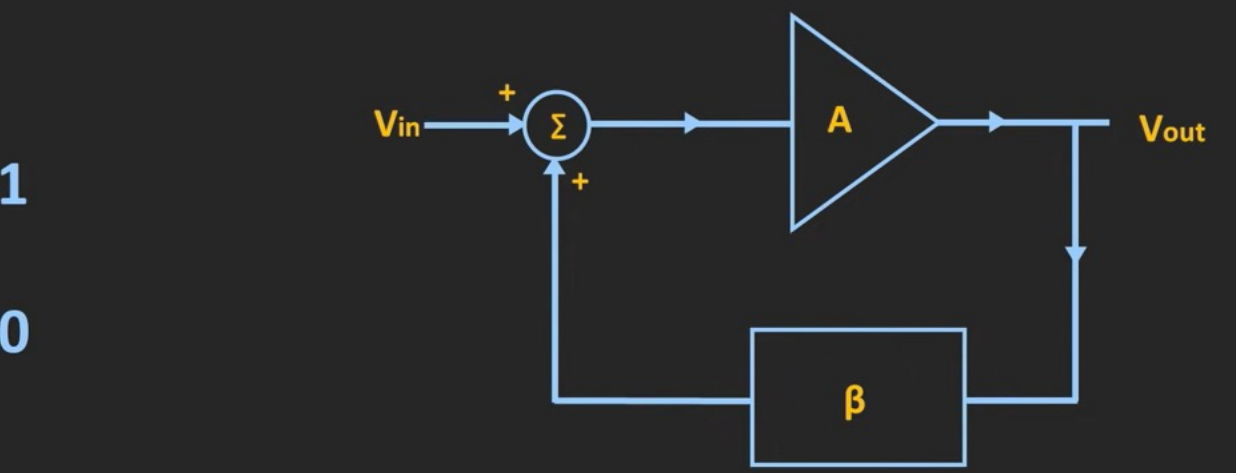

Aβ <1

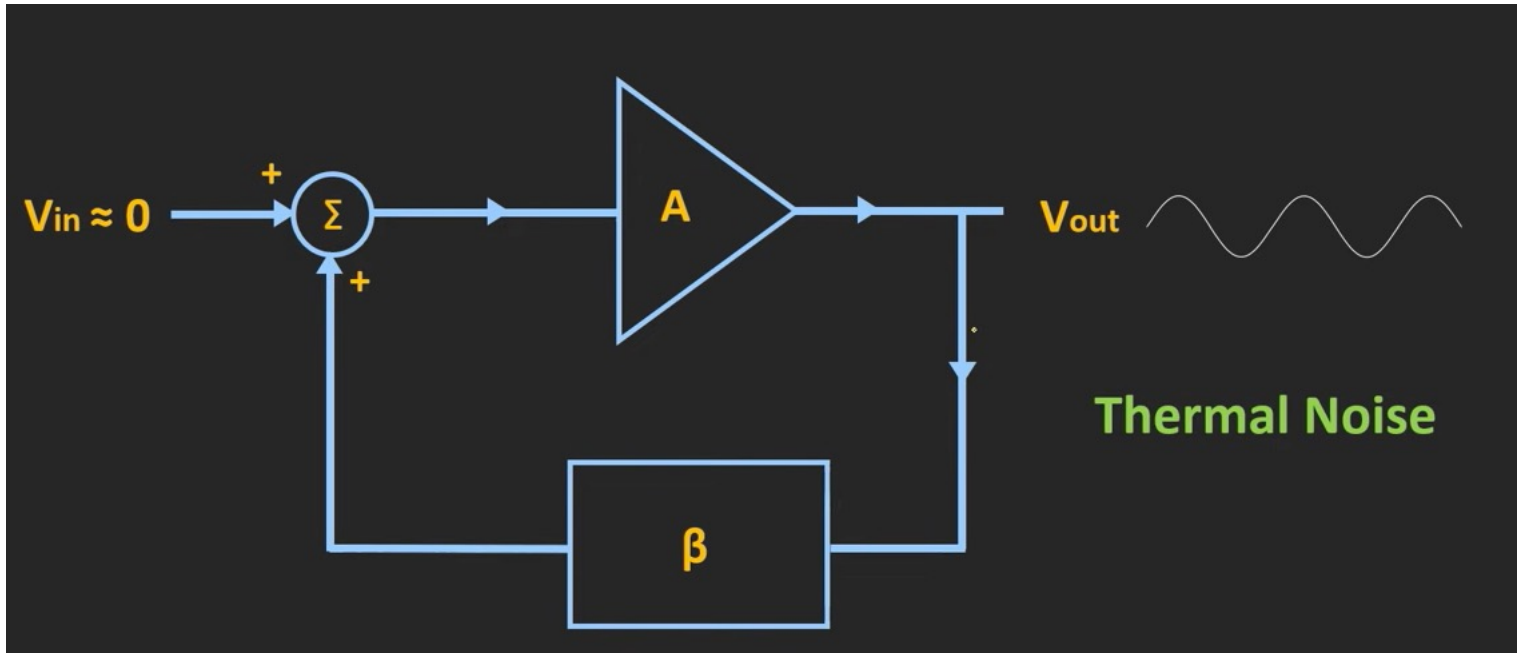

Aβ >1

Barkhausen Criteria

$|A\beta| = 1$

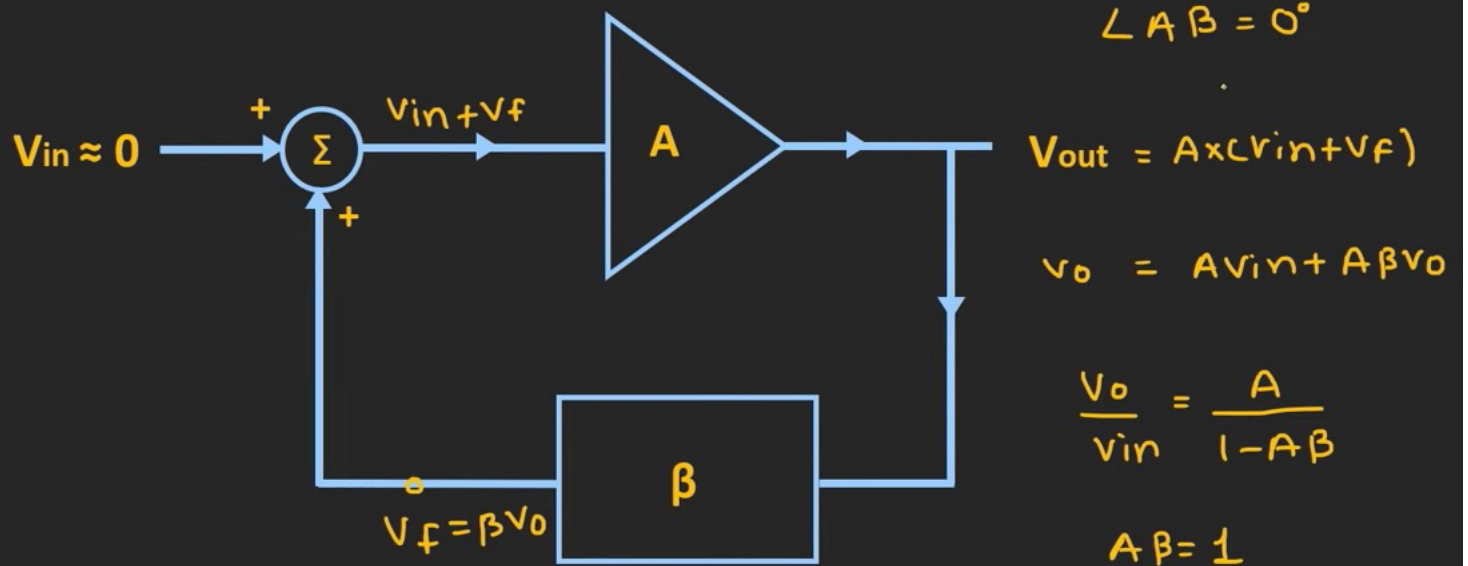
$\angle A\beta = 0$





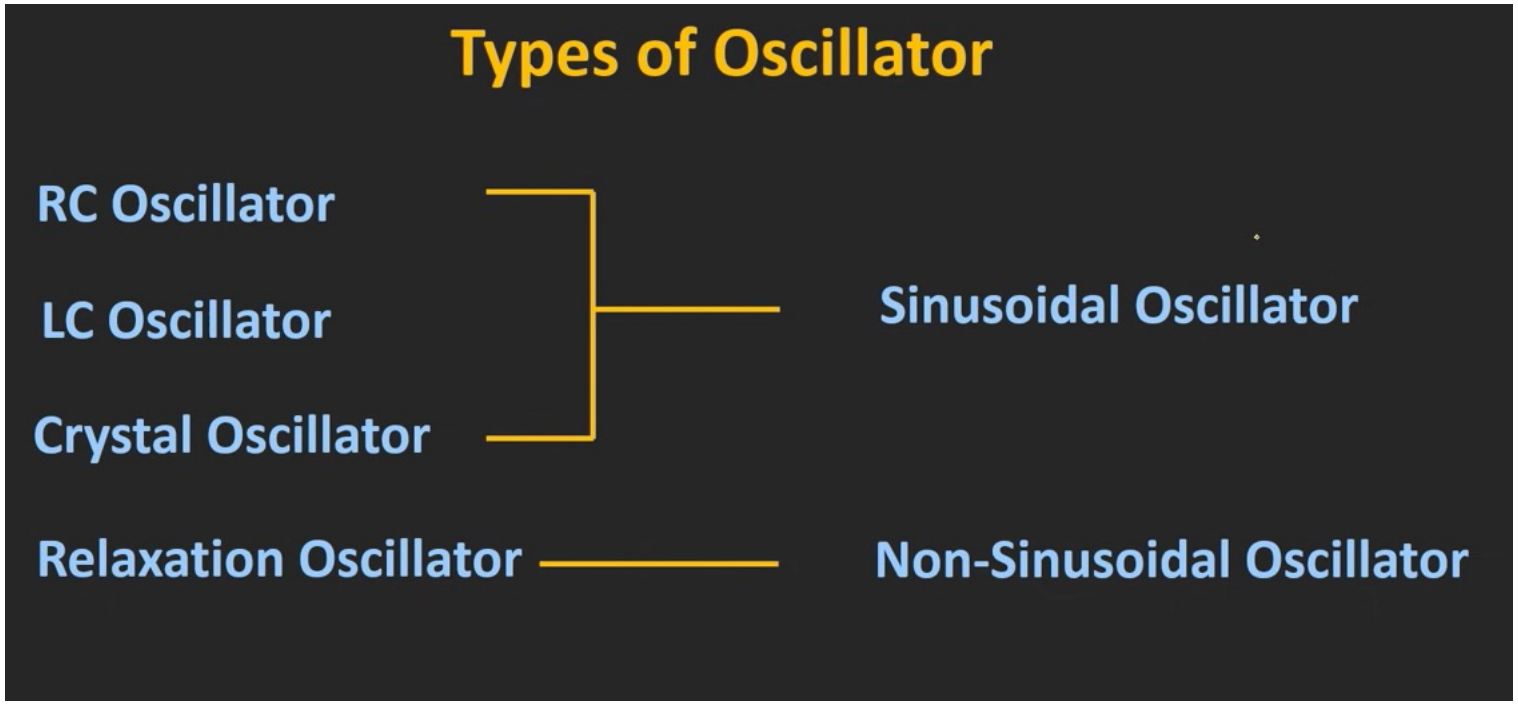
Frequency selective circuit; only one particular freq will have phase shift=0 while other freqs will have a diff phase.

Oscillator

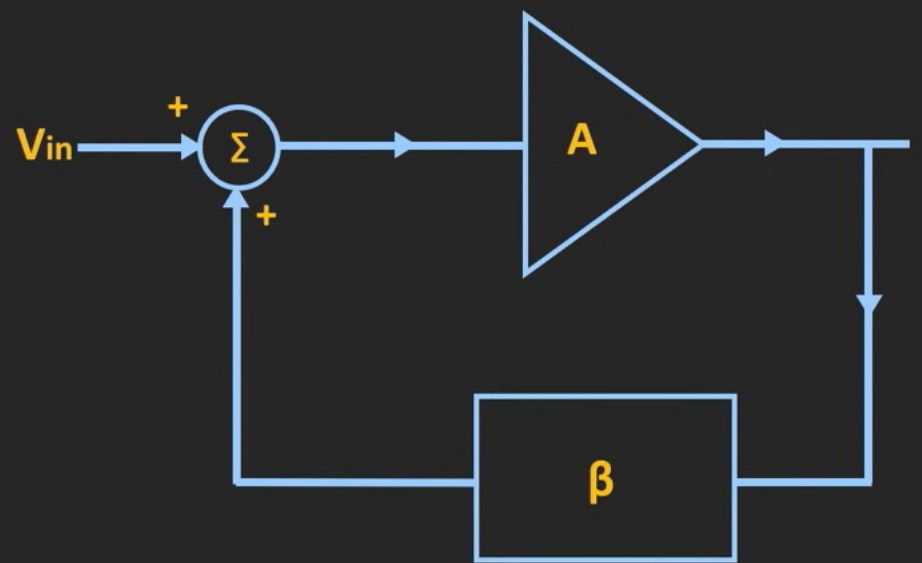


$|A\beta| = 1$

$\angle A\beta = 0^\circ$

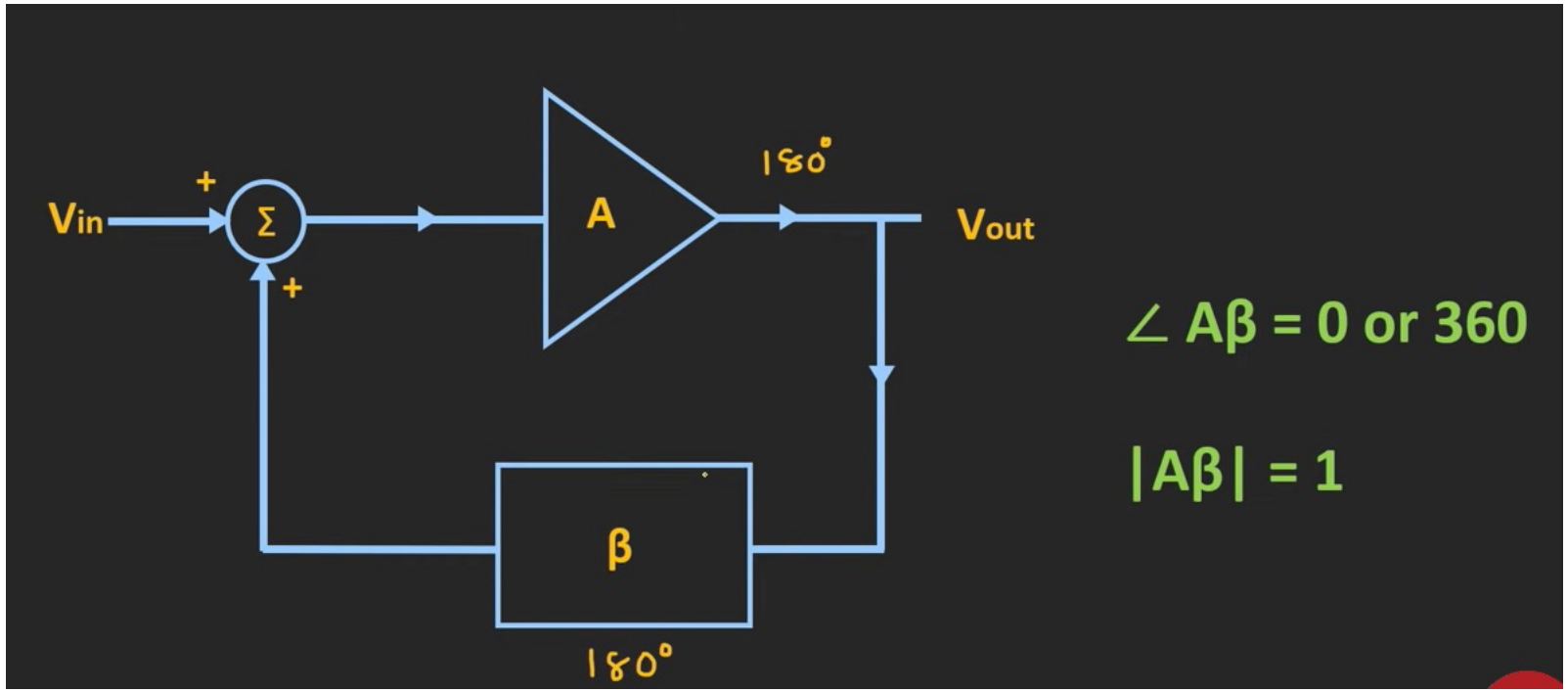

Types of Oscillator
RC Oscillator
LC Oscillator
Crystal Oscillator
Sinusoidal Oscillator
Relaxation Oscillator
Non-Sinusoidal Oscillator

RC Phase Shift Oscillator

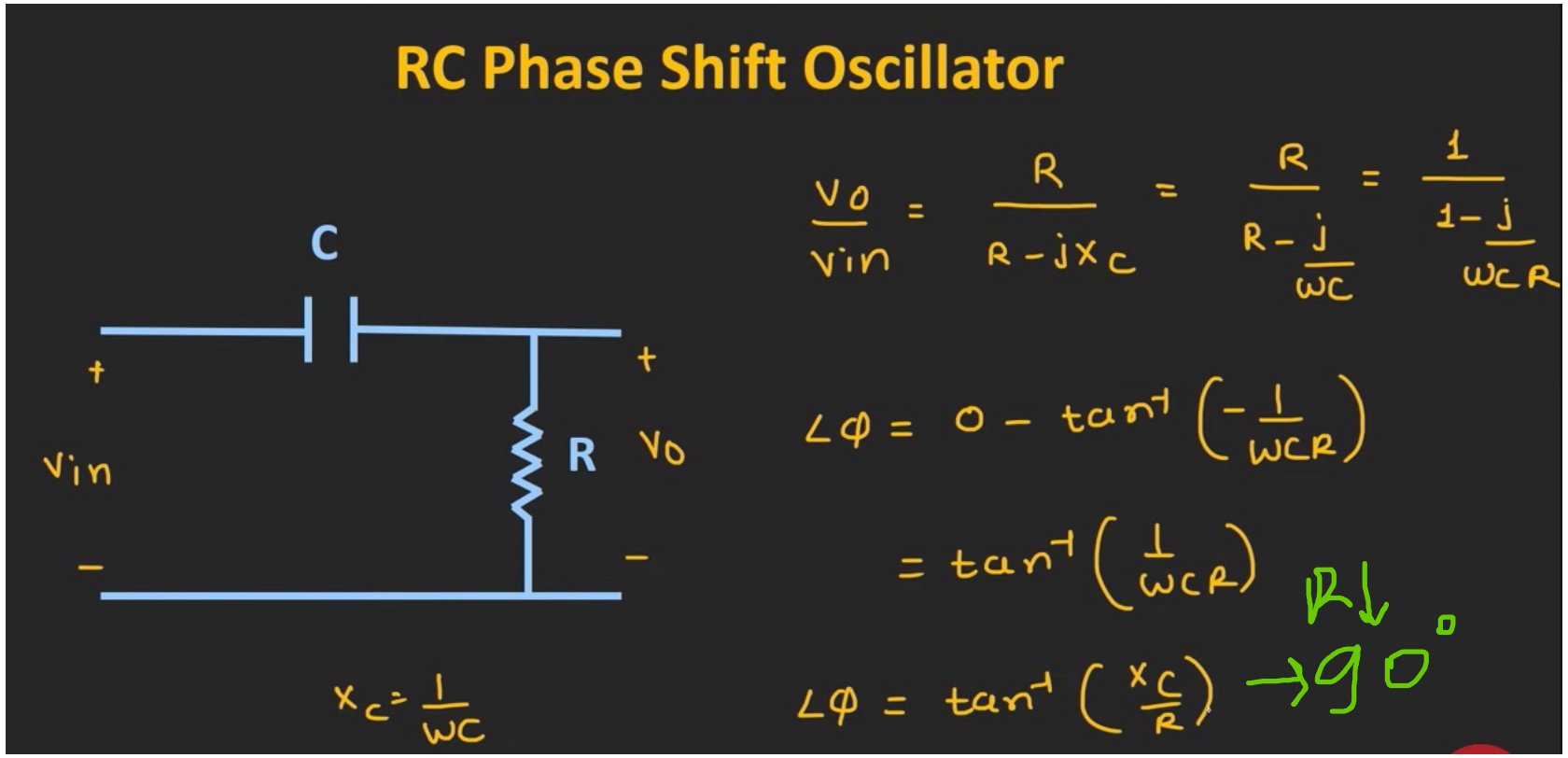


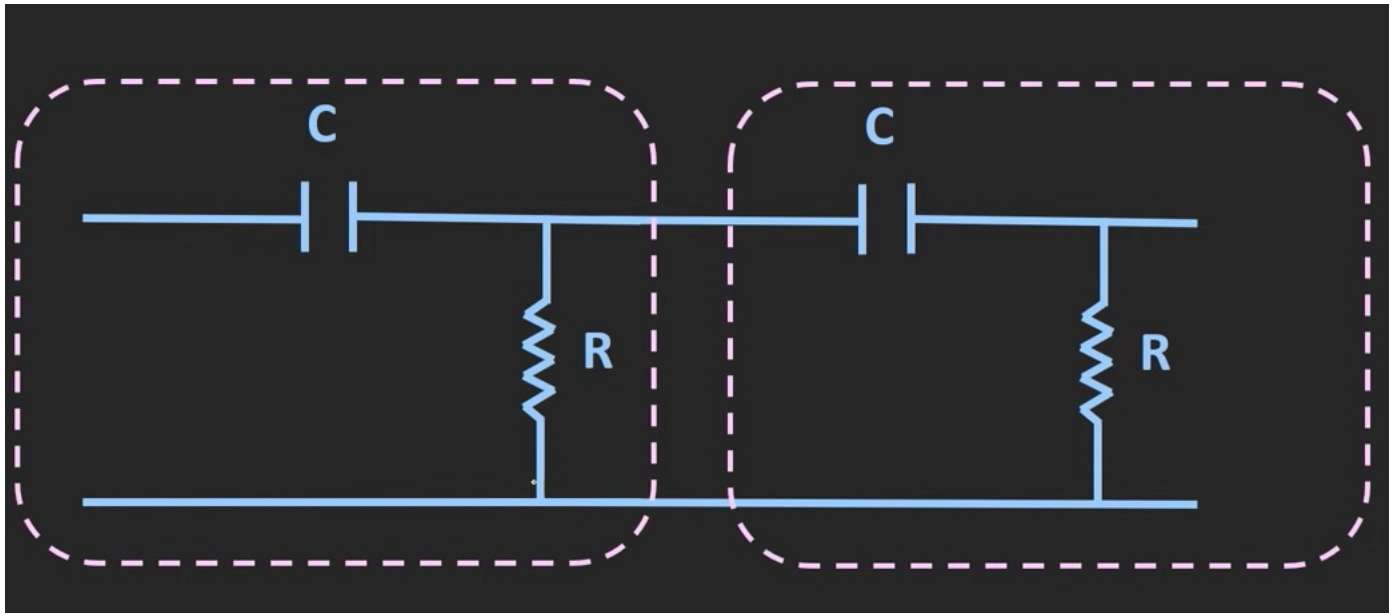
Low Frequencies

Hundreds of kHZ


Vin
+
Σ
+
A
180°
Vout
β
180°
∠ Aβ = 0 or 360
|Aβ| = 1

HPF or Phase lead circuit

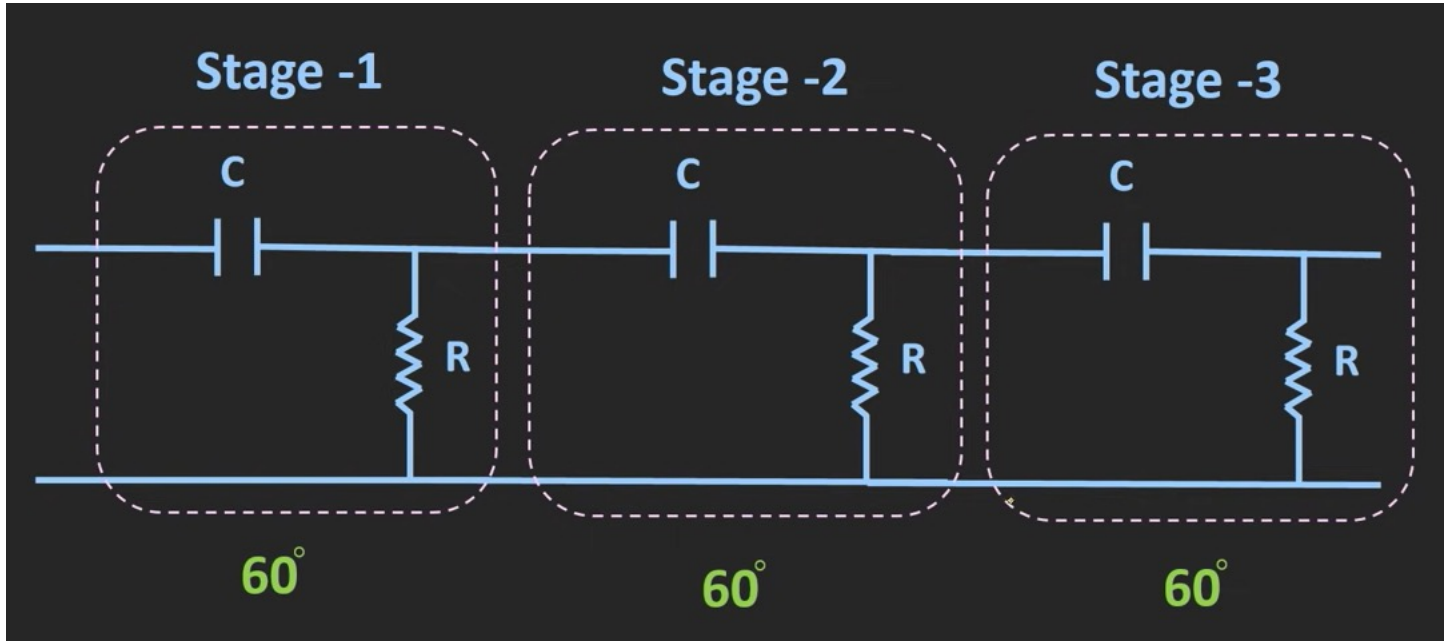


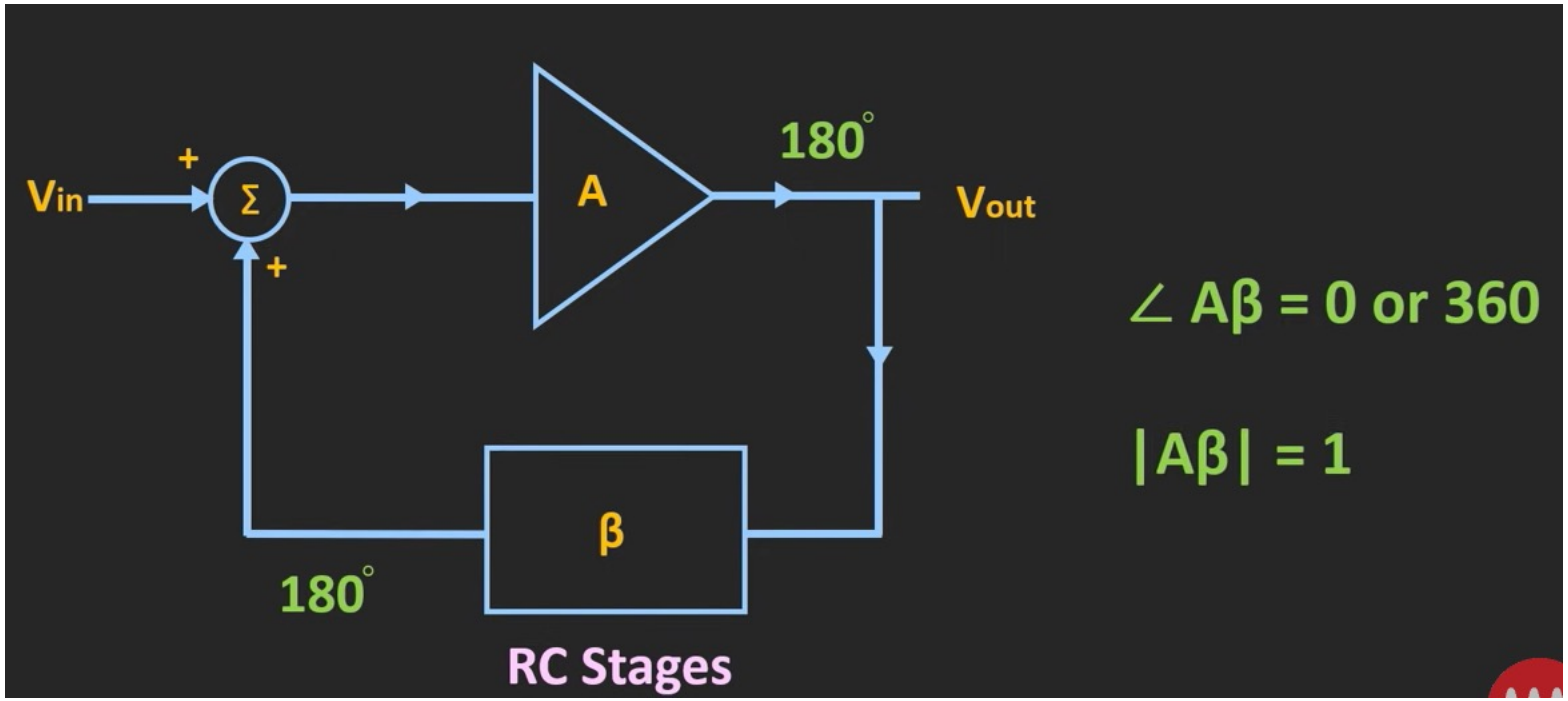


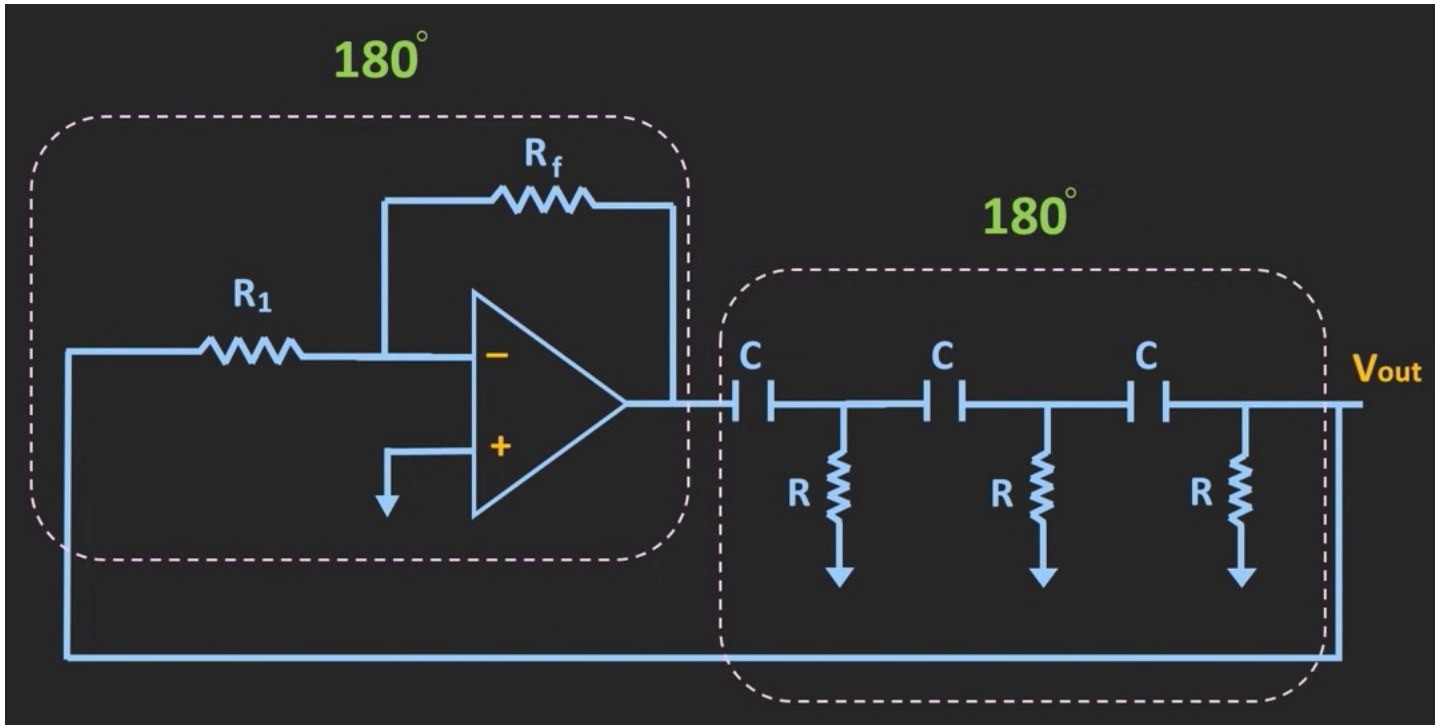
$R = 1\Omega \simeq 0$

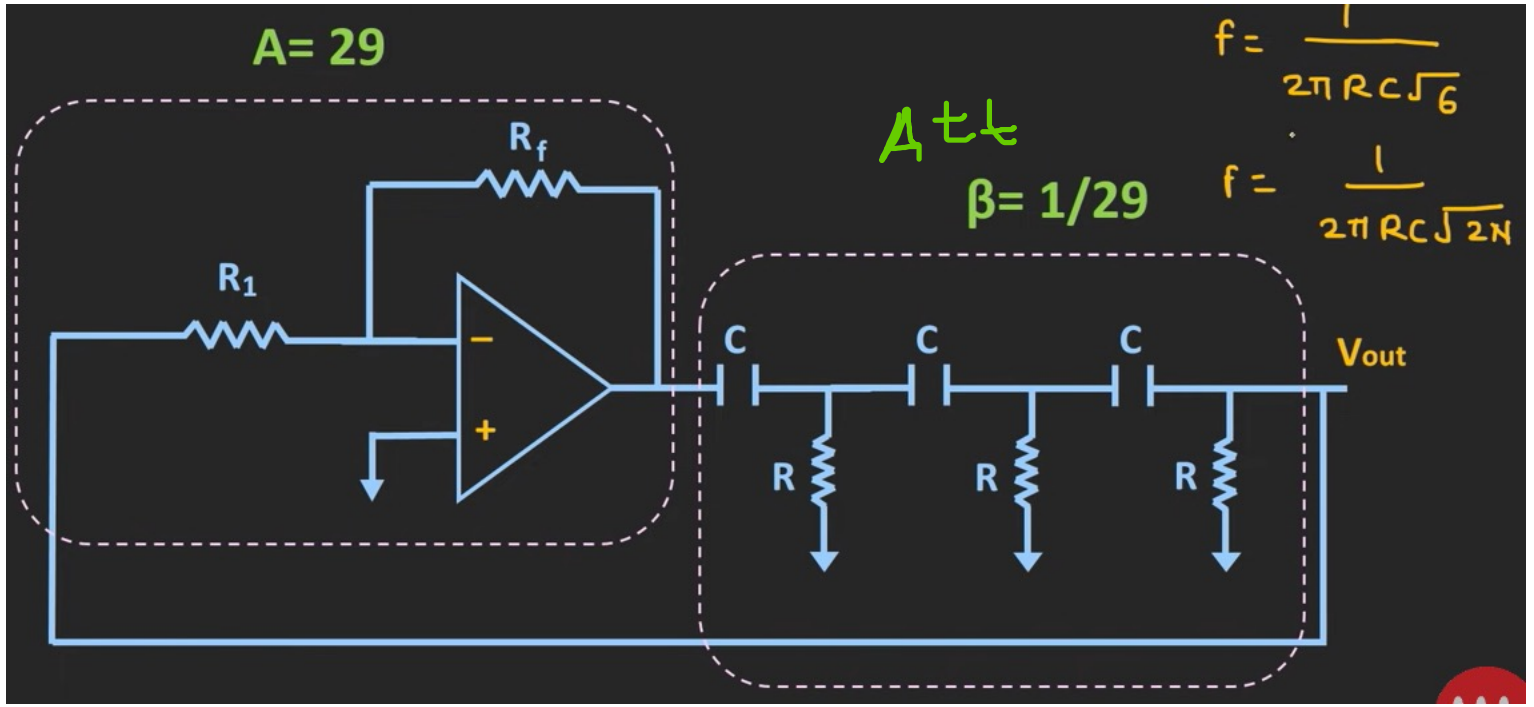
Gain $\rightarrow 0$

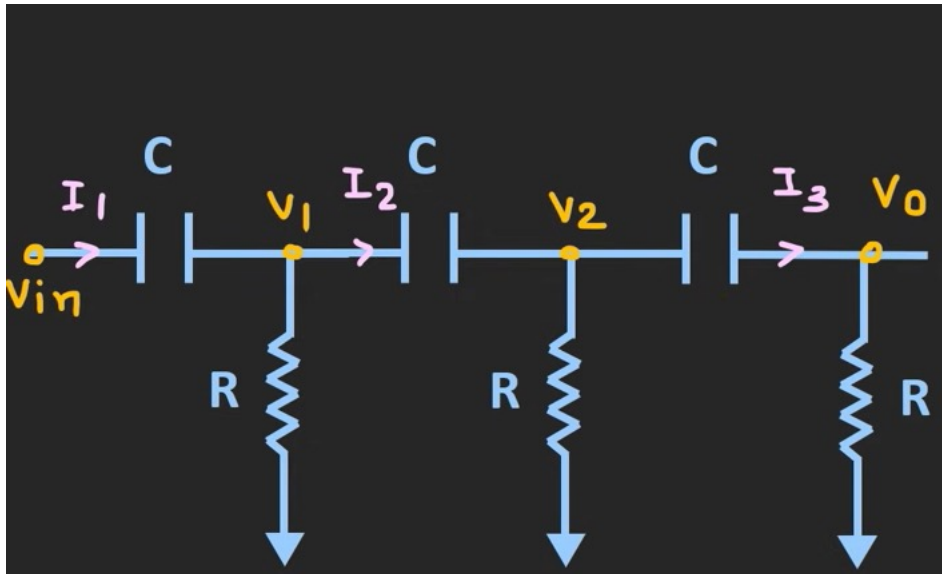
For stable phase shift




Vin
+
Σ
+
A
180°
Vout
β
180°
RC Stages
∠ Aβ = 0 or 360
|Aβ| = 1


180°
R_f
R_1
−
+
180°
C
C
C
R
R
R
Vout


A= 29
R_f
R_1
−
+
C
C
C
R
R
R
Vout
A ≥ 29
β= 1/29
$f = \frac{1}{2\pi RC\sqrt{6}}$
$f = \frac{1}{2\pi RC\sqrt{2N}}$

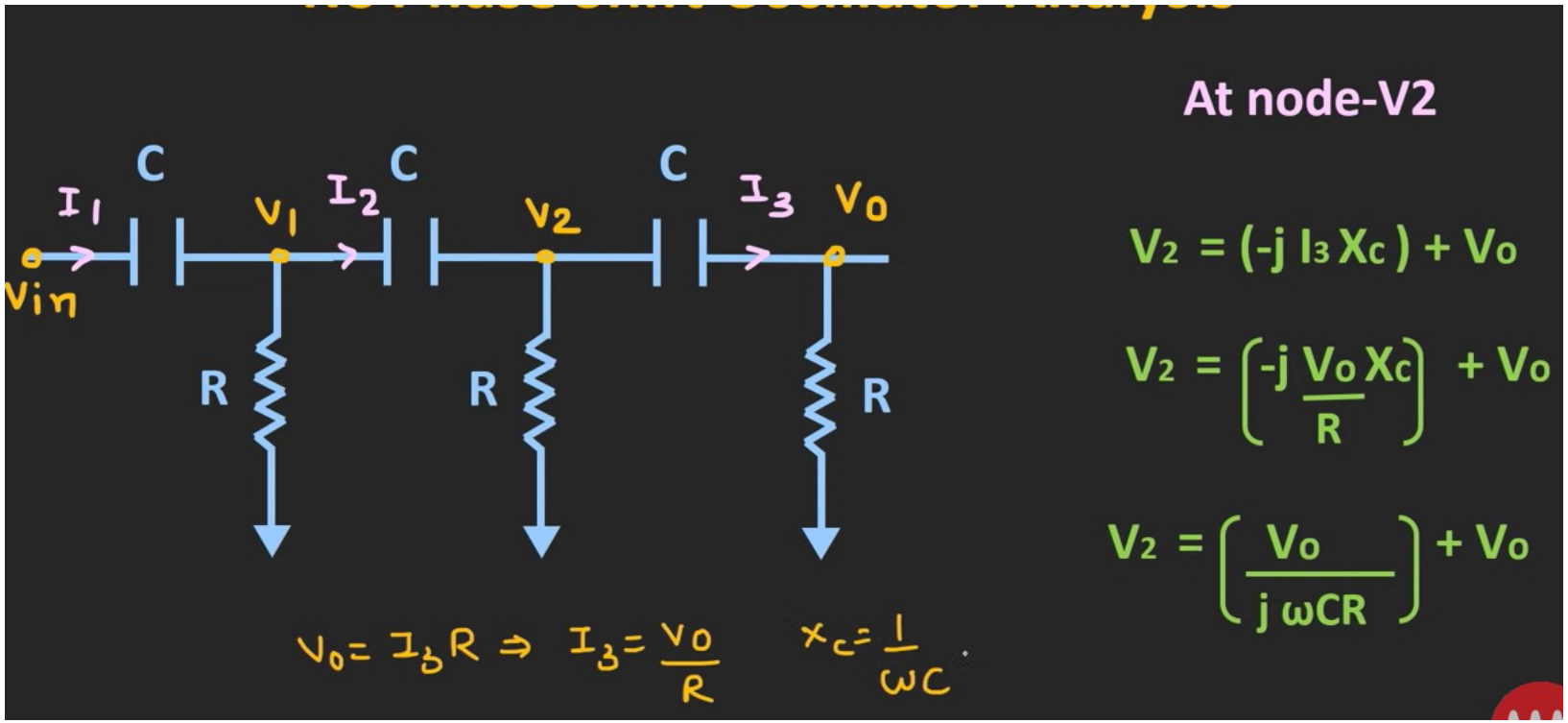


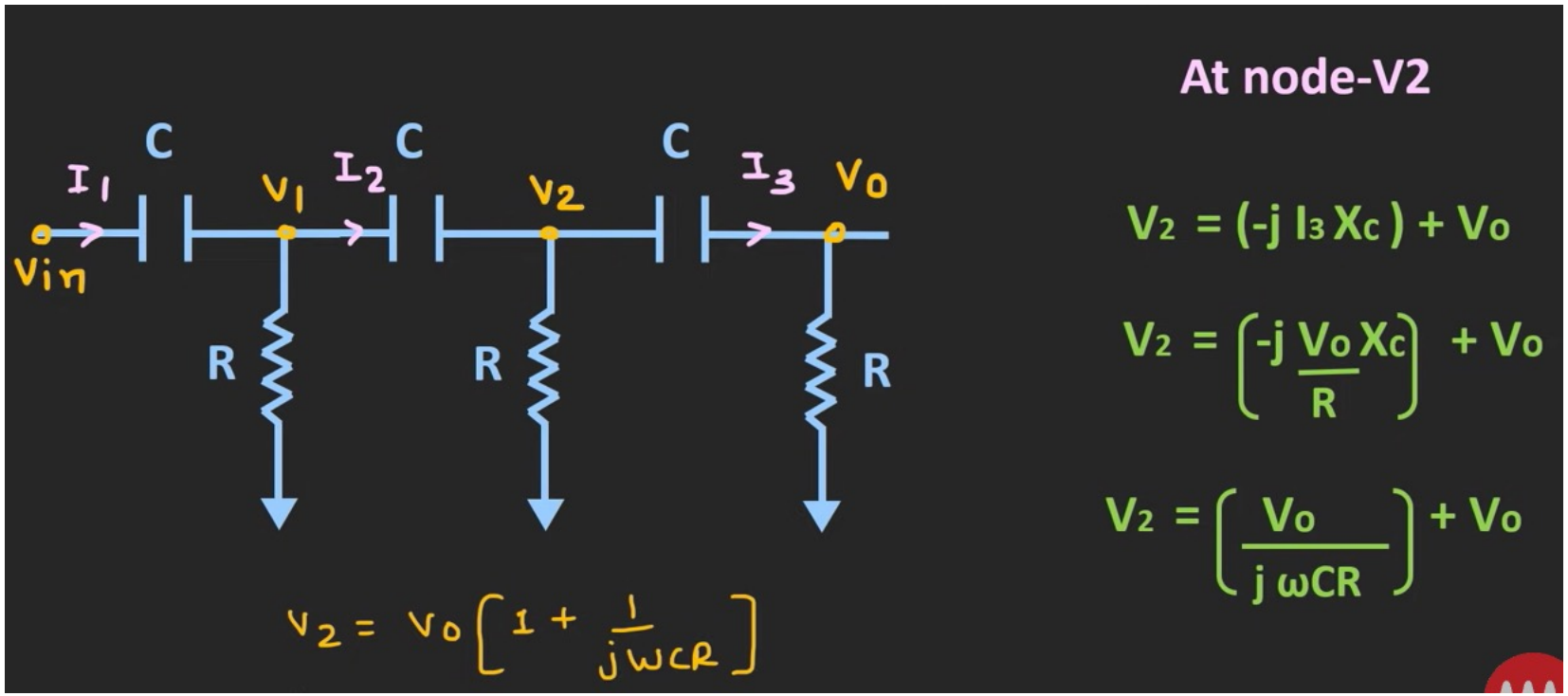
At node-V2

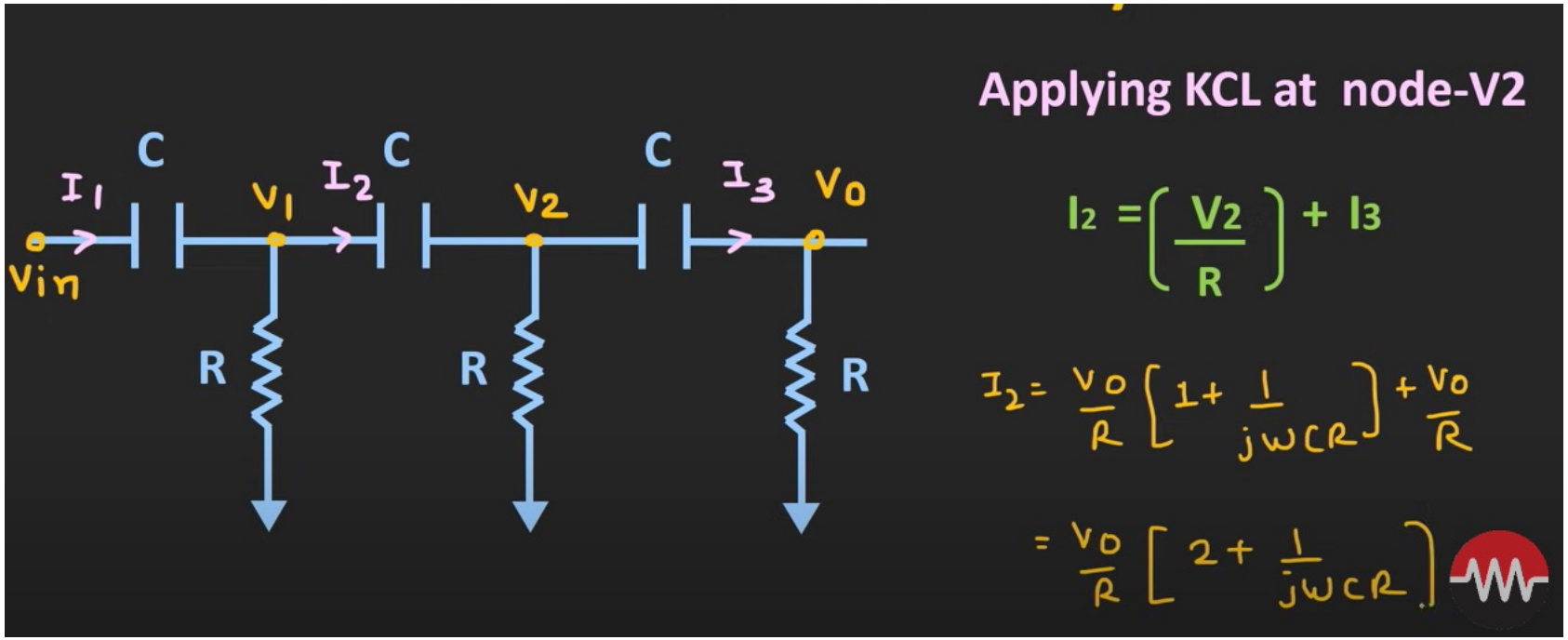
$$V_2 = (-j\, I_3 X_C) + V_o$$

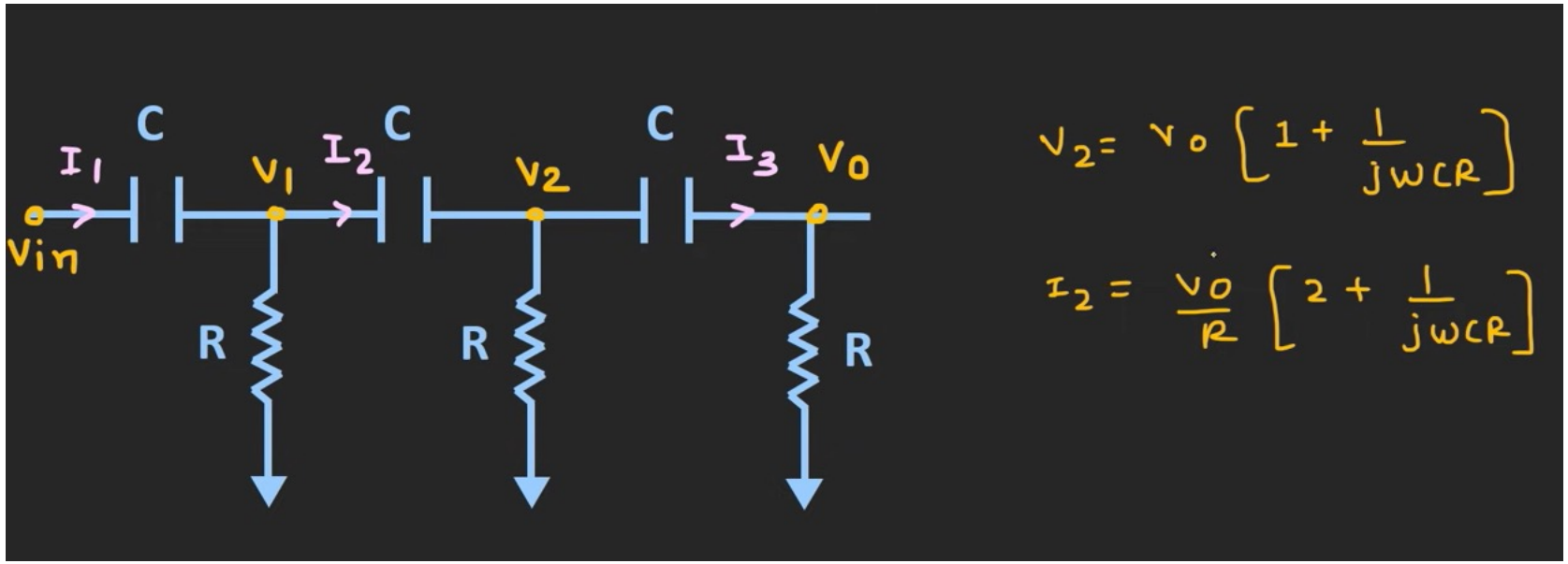
$$V_2 = \left(-j \frac{V_o X_C}{R}\right) + V_o$$

$$V_2 = \left(\frac{V_o}{j\,\omega CR}\right) + V_o$$


C
C
C
I1
I2
I3
V1
V2
Vo
Vin
R
R
R
Vo = I3 R ⇒ I3 = Vo / R
Xc = 1 / ωC
At node-V2
V2 = (-j I3 Xc) + Vo
V2 = (-j Vo Xc / R) + Vo
V2 = (Vo / j ωCR) + Vo

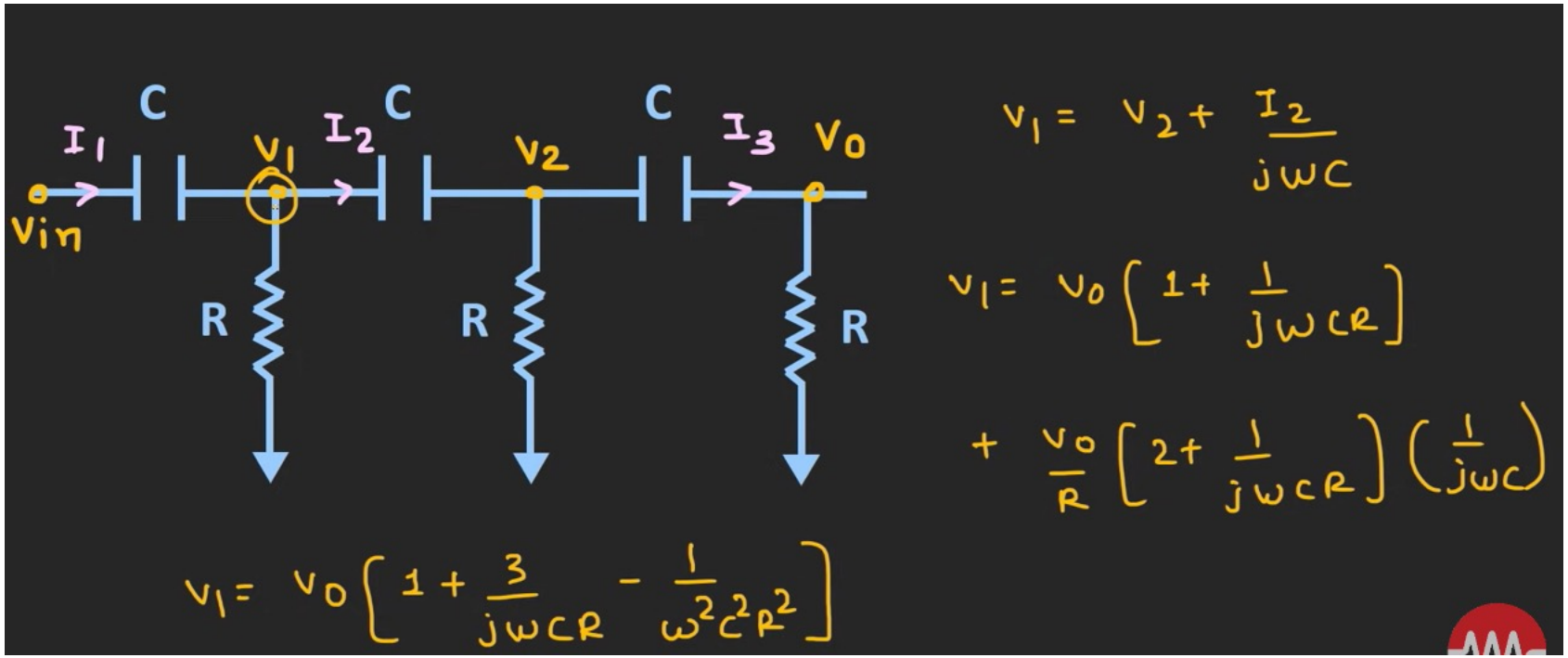

C
C
C
I1
I2
I3
V1
V2
Vo
Vin
R
R
R
At node-V2
V2 = (-j I3 Xc) + Vo
V2 = (-j Vo Xc / R) + Vo
V2 = (Vo / j ωCR) + Vo
V2 = Vo [1 + 1 / jωCR]

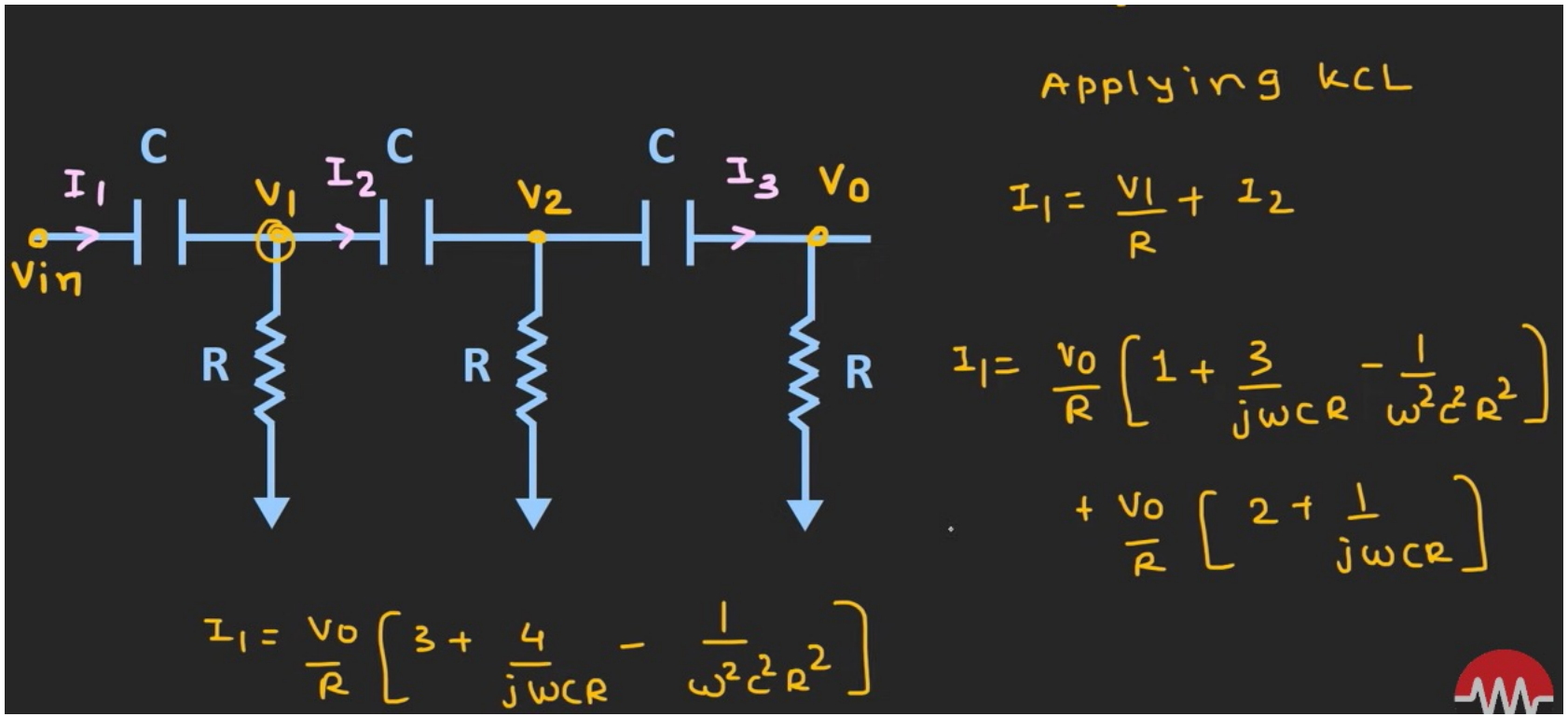

C
C
C
I1
I2
I3
V1
V2
Vo
Vin
R
R
R
Applying KCL at node-V2
I2 = (V2/R) + I3
I2 = Vo/R [1 + 1/jωCR] + Vo/R
= Vo/R [2 + 1/jωCR]

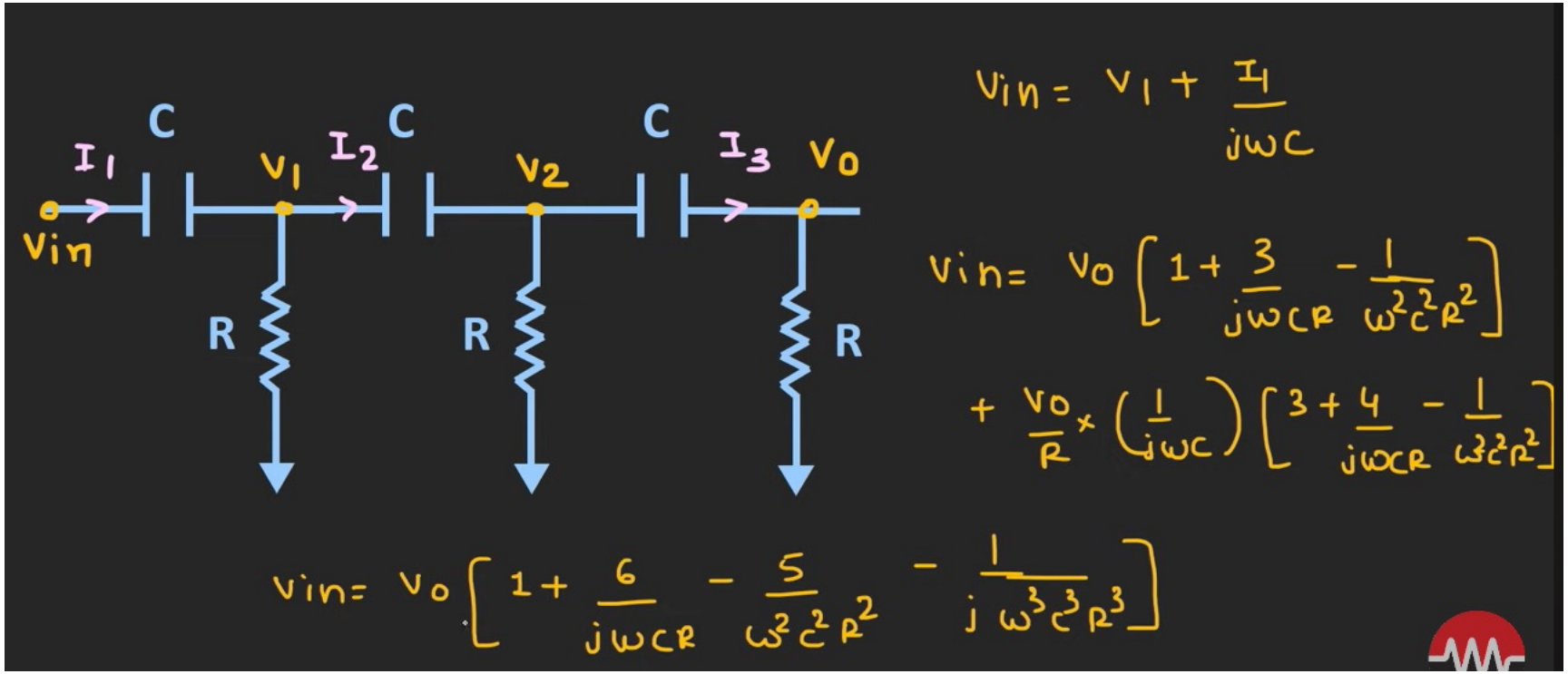


$$V_2 = V_o \left[1 + \frac{1}{j\omega CR}\right]$$

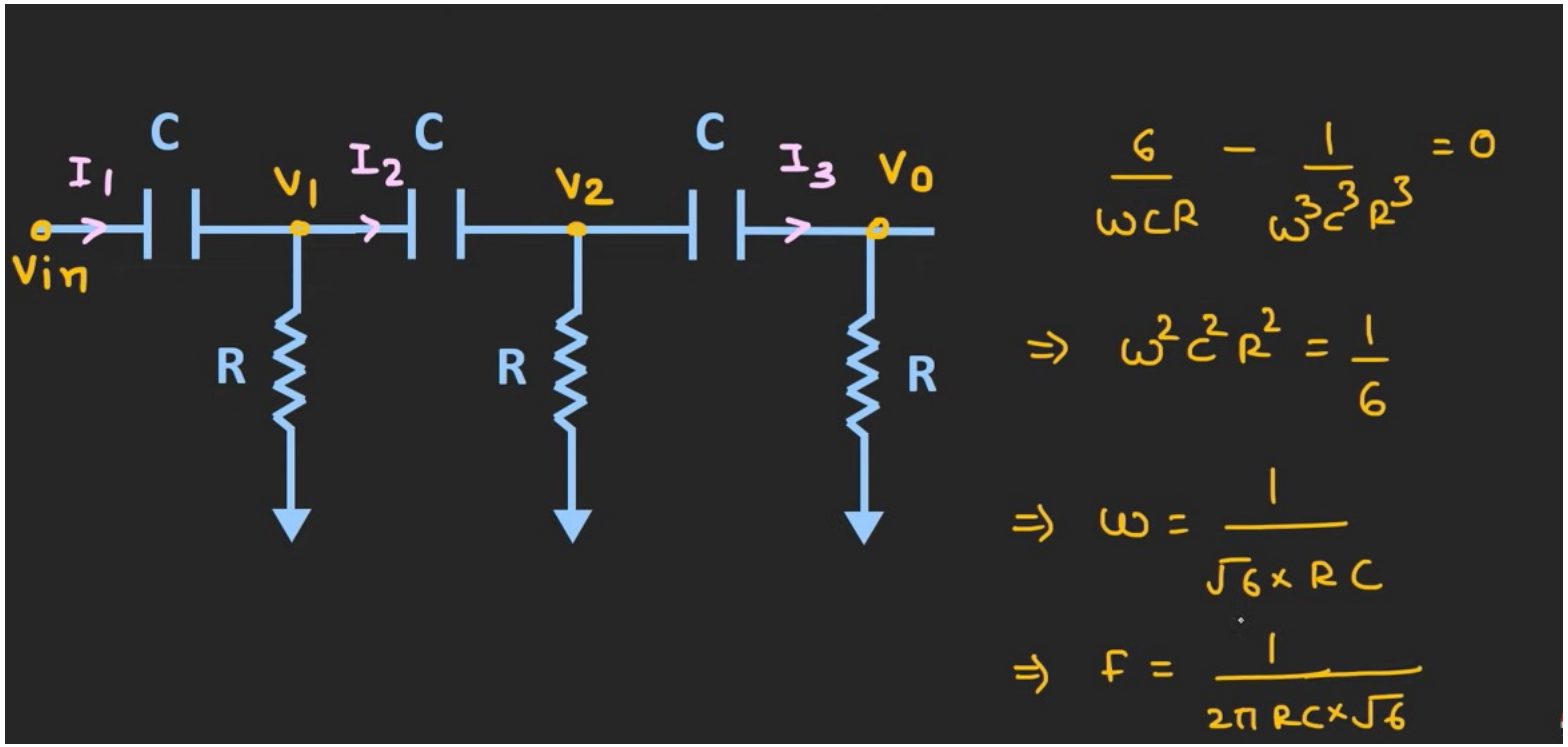
$$I_2 = \frac{V_o}{R} \left[2 + \frac{1}{j\omega CR}\right]$$


C
C
C
I_1
I_2
I_3
V_1
V_2
V_0
V_{in}
R
R
R
$V_1 = V_2 + \frac{I_2}{j\omega C}$
$V_1 = V_0\left[1 + \frac{1}{j\omega CR}\right]$
$+ \frac{V_0}{R}\left[2 + \frac{1}{j\omega CR}\right]\left(\frac{1}{j\omega C}\right)$
$V_1 = V_0\left[1 + \frac{3}{j\omega CR} - \frac{1}{\omega^2 C^2 R^2}\right]$


C
C
C
I_1
I_2
I_3
V_1
V_2
V_0
Vin
R
R
R
Applying KCL
$I_1 = \frac{V_1}{R} + I_2$
$I_1 = \frac{V_0}{R}\left[1 + \frac{3}{j\omega CR} - \frac{1}{\omega^2 C^2 R^2}\right]$
$+ \frac{V_0}{R}\left[2 + \frac{1}{j\omega CR}\right]$
$I_1 = \frac{V_0}{R}\left[3 + \frac{4}{j\omega CR} - \frac{1}{\omega^2 C^2 R^2}\right]$



Output voltage will only have a real part; imaginary part of this expression = 0



$$\frac{6}{\omega CR} - \frac{1}{\omega^3 C^3 R^3} = 0$$

$$\Rightarrow \omega^2 C^2 R^2 = \frac{1}{6}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{6} \times RC}$$

$$\Rightarrow f = \frac{1}{2\pi RC \times \sqrt{6}}$$

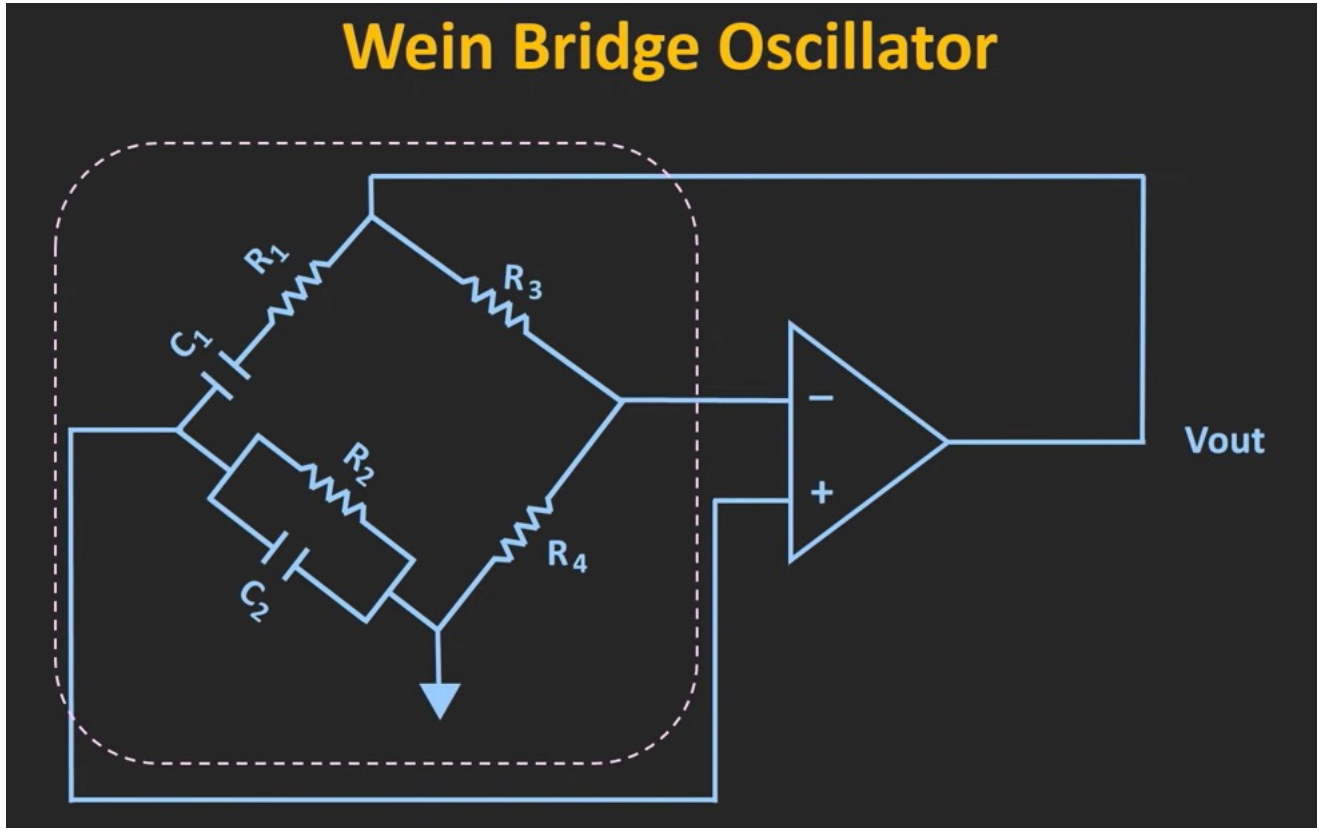
$$V_{in} = V_o \left[1 - \frac{5}{\omega^2 C^2 R^2} \right]$$

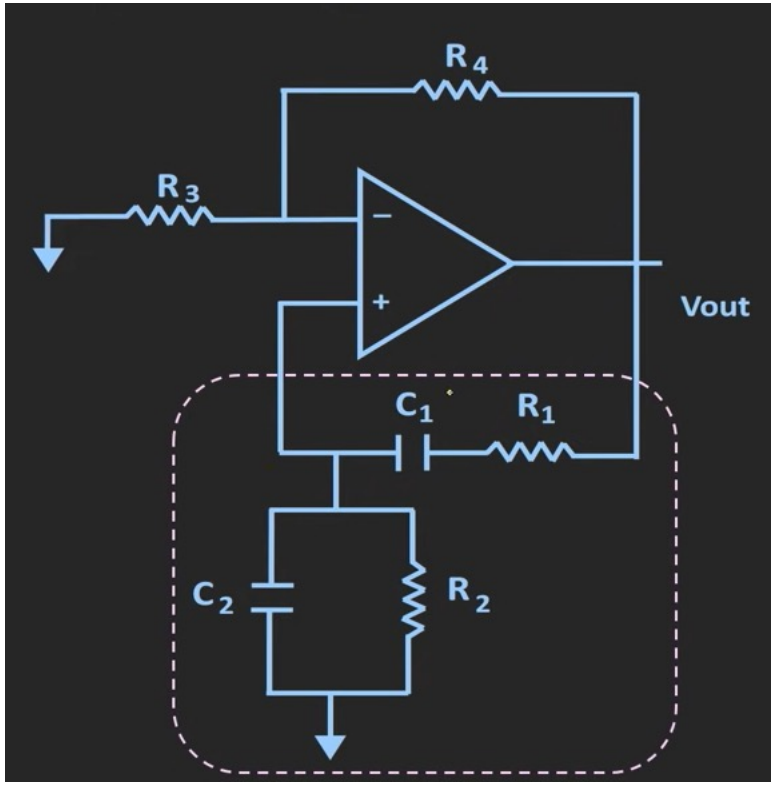
$$\omega = \frac{1}{\sqrt{6} \times RC}$$

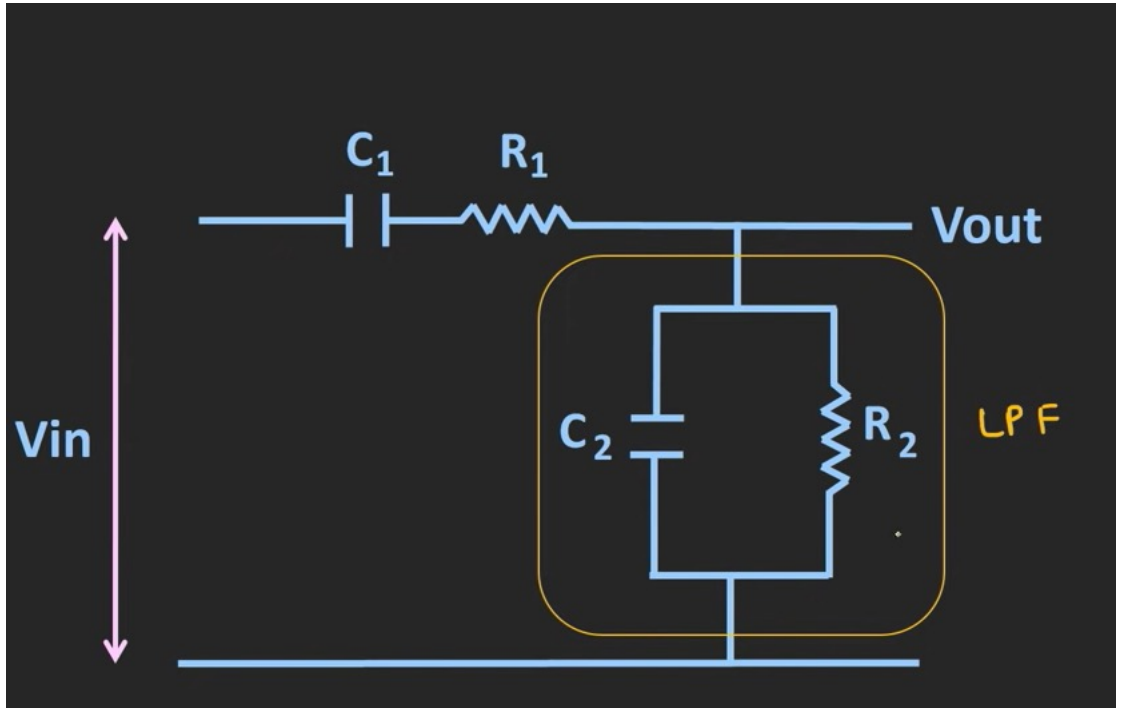
$$V_{in} = V_o \left[1 - \frac{5}{\frac{1}{6R^2C^2} \times R^2C^2} \right]$$

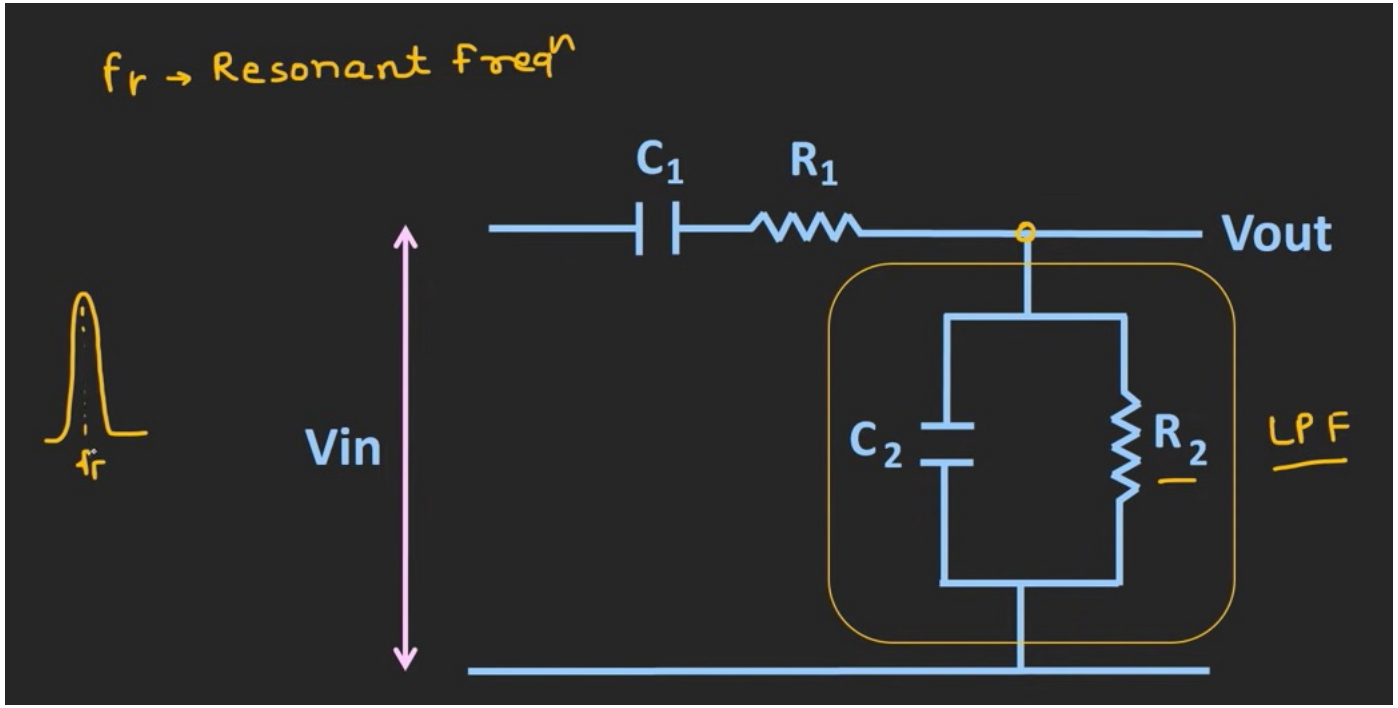
$$V_{in} = V_o \left[1 - 30 \right]$$

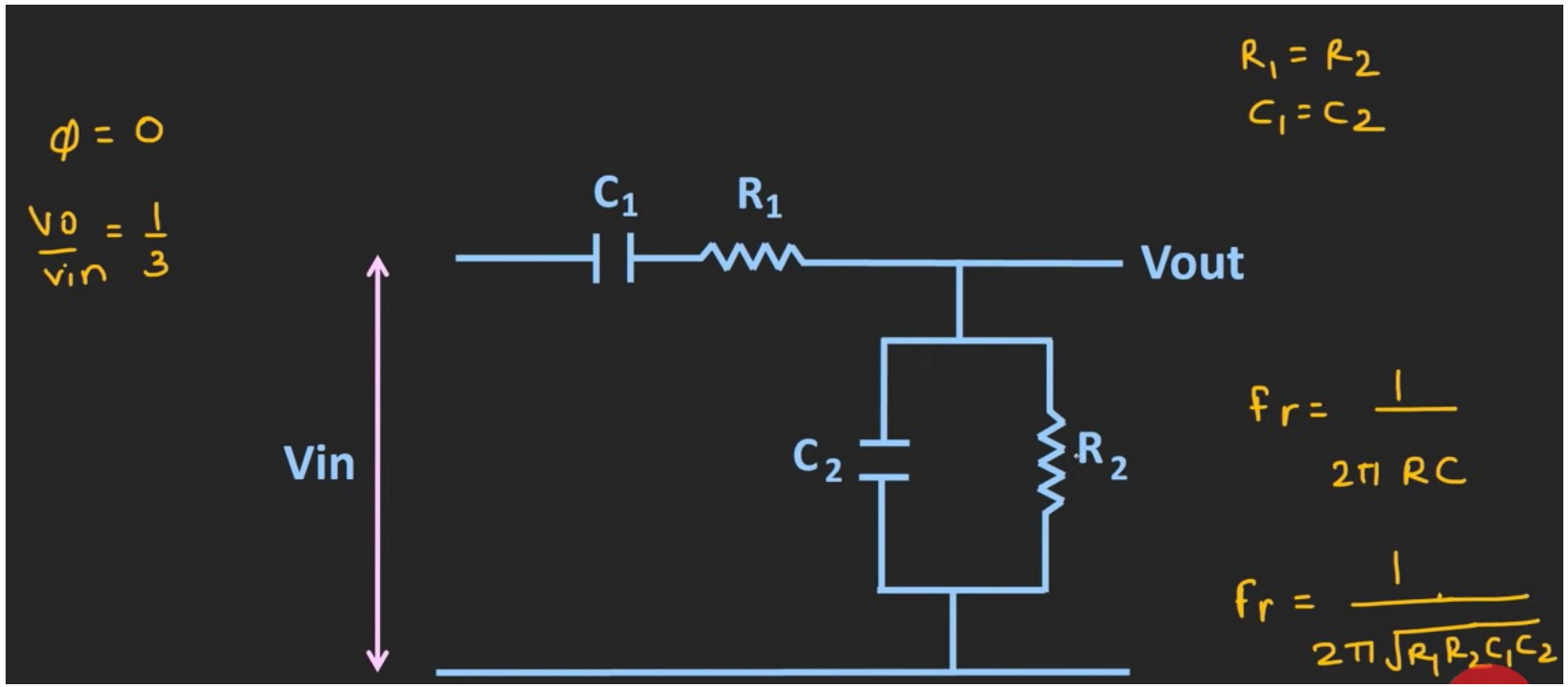
$$\Rightarrow \boxed{\frac{V_o}{V_{in}} = \frac{-1}{29}}$$

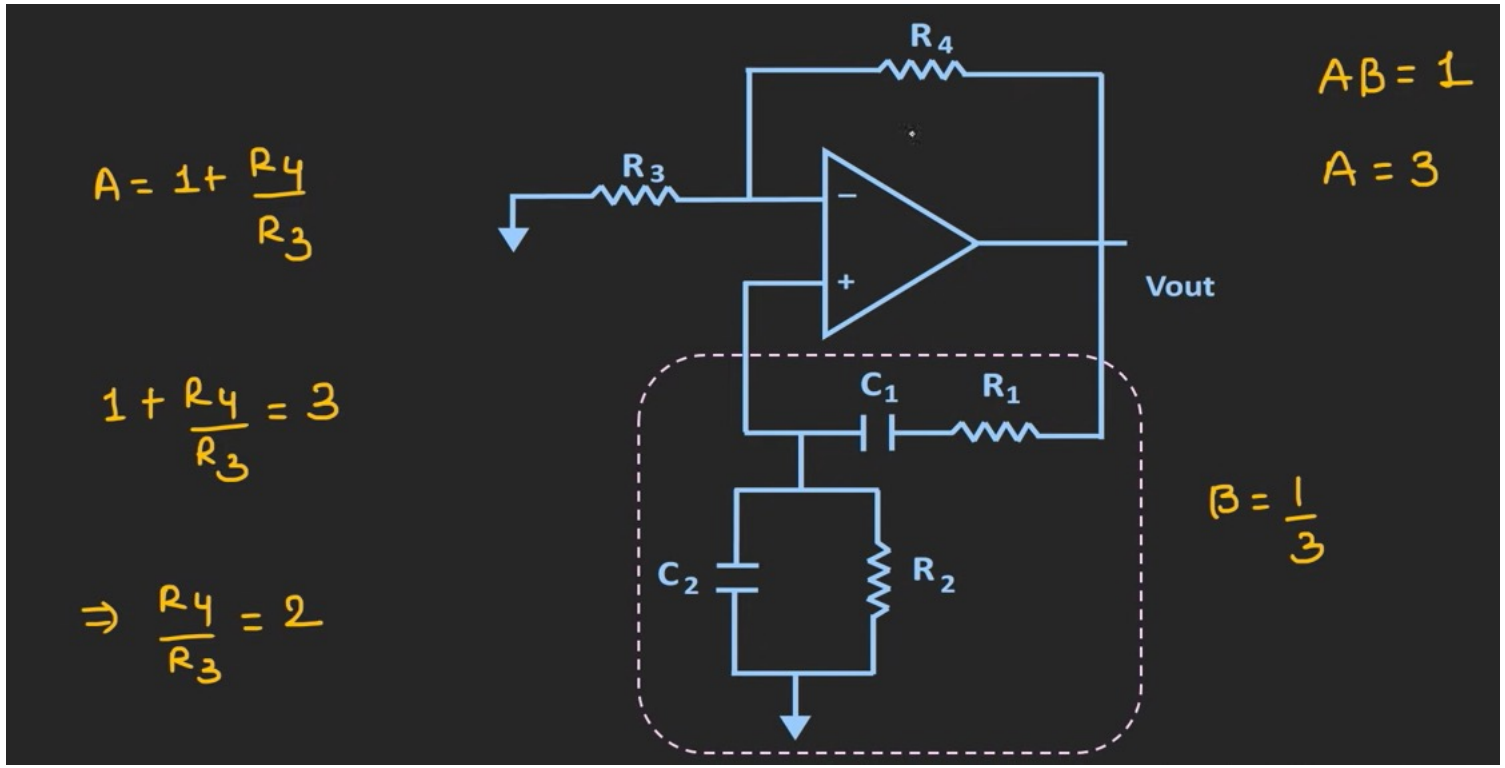

Wein Bridge Oscillator
R1
C1
R3
R2
C2
R4
Vout


R4
R3
Vout
C1
R1
C2
R2


C1
R1
Vout
Vin
C2
R2
LP F


fr → Resonant freq^n
C1
R1
Vout
Vin
fr
C2
R2
LP F


$\phi = 0$
$\frac{V_0}{V_{in}} = \frac{1}{3}$
$R_1 = R_2$
$C_1 = C_2$
C_1
R_1
Vout
Vin
C_2
R_2
$f_r = \frac{1}{2\pi RC}$
$f_r = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$


$A = 1 + \frac{R_4}{R_3}$
$1 + \frac{R_4}{R_3} = 3$
$\Rightarrow \frac{R_4}{R_3} = 2$
$A\beta = 1$
$A = 3$
$\beta = \frac{1}{3}$
R4
R3
Vout
C1
R1
C2
R2

$R_1 = R_2$

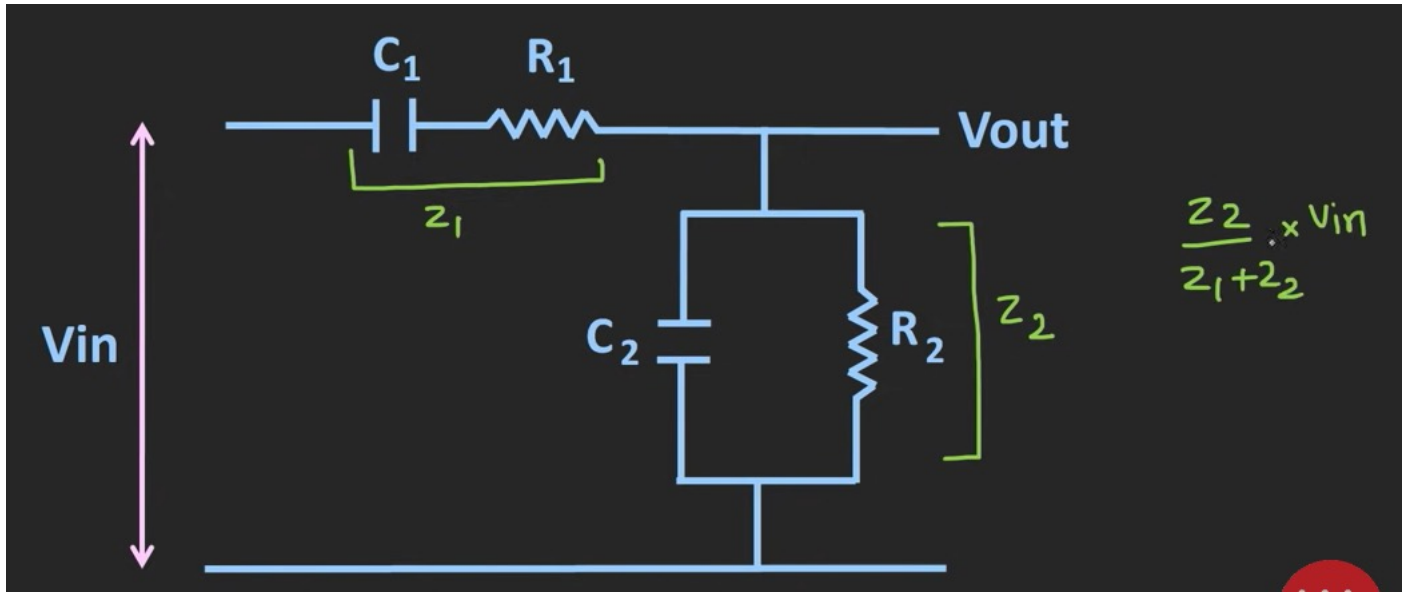
$C_1 = C_2$

$f = 10\,kHz$

$C = 0.01\,\mu F$

$$f_r = \frac{1}{2\pi R C}$$

$$R = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 10^{4} \times 10^{-8}} \simeq 1.59\,k\Omega.$$


C1
R1
Vout
Z1
Vin
C2
R2
Z2
Z2 / Z1+Z2 . x Vin

$$\frac{V_o}{V_{in}} = \frac{Z_2}{Z_1 + Z_2}$$

$$Z_2 = R_2 \,||\, \left(\frac{1}{j\omega C_2}\right) = \frac{R_2}{1 + j\omega R_2 C_2}$$

$$\frac{V_o}{V_{in}} = \frac{\dfrac{R_2}{1 + j\omega R_2 C_2}}{R_1 + \dfrac{1}{j\omega C_1} + \dfrac{R_2}{1 + j\omega C_2 R_2}}$$

$$\frac{V_o}{V_{in}} = \frac{R_2 (j\omega C_1)}{R_1 (j\omega C_1)(1 + j\omega C_2 R_2) + 1 + j\omega C_2 R_2 + j\omega R_2 C_1}$$

$$\frac{V_o}{V_{in}} = \frac{j\omega R_2 C_1}{\underbrace{1-\omega^2 R_1 R_2 C_1 C_2}_{0} + j\omega\left[R_1 C_1 + R_2 C_2 + R_2 C_1\right]}$$

$$\omega^2 R_1 R_2 C_1 C_2 = 1$$

$$\Rightarrow \omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}} \Rightarrow f = \frac{1}{2\pi \times \sqrt{R_1 R_2 C_1 C_2}} \Rightarrow f = \frac{1}{2\pi RC}$$

$$\frac{V_o}{V_{in}} = \frac{R_2 C_1}{R_1 C_1 + R_2 C_2 + R_2 C_1} = \beta \qquad \beta = \frac{1}{3}$$

$$A\beta = 1$$

$$A = \frac{1}{\beta} = \frac{R_1 C_1 + R_2 C_2 + R_2 C_1}{R_2 C_1} = \frac{R_1}{R_2} + \frac{C_2}{C_1} + 1$$

$$A = 1 + \frac{R_4}{R_3} = \frac{R_1}{R_2} + \frac{C_2}{C_1} + 1$$

$$\Rightarrow \frac{R_4}{R_3} = \frac{R_1}{R_2} + \frac{C_2}{C_1}$$

$$R_1 = R_2$$

$$C_1 = C_2$$

$$\Rightarrow \frac{R_4}{R_3} = 2$$